

IILM UNIVERSITY

Greater Noida, Uttar Pradesh

INTRODUCTION TO ARTIFICIAL INTELLIGENCE

An Internship Presentation Report on the Foundations, Reasoning Methods, and Applications of Artificial Intelligence

INTERNSHIP ORGANIZATION

Kodacy

DOMAIN

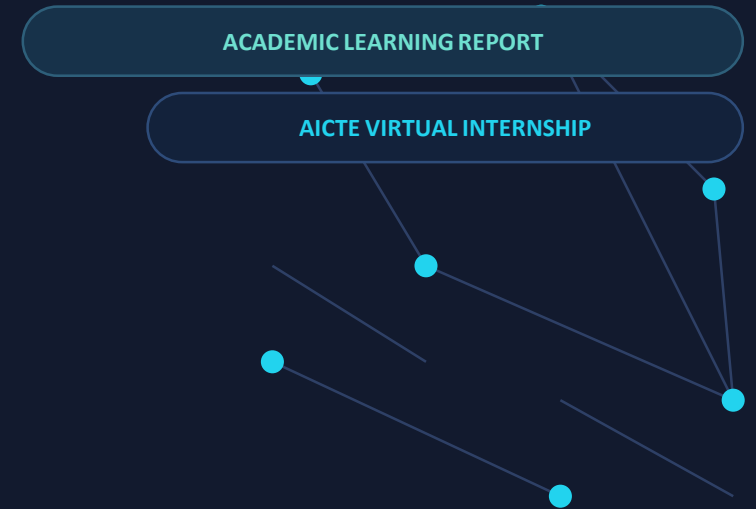
Introduction to Artificial Intelligence

PRESENTED BY

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COMPLETED ON

10th September, 2026



Candidate & Internship Overview

Summary of candidate details, internship engagement, and the core learning track.



Candidate & Track Details

Candidate Name:	Ayush Jha
Roll No.:	2410031188
Section:	3CSE24
Host Organization:	Kodacy
Domain:	Introduction to Artificial Intelligence
Internship Mode:	AICTE Virtual Mode
Institution:	IILM University, Greater Noida



Internship Focus Areas

- Foundations of Artificial Intelligence & Intelligent Agents
- Problem-Solving through Search & Heuristics
- Knowledge Representation & Logical Reasoning
- Probability-Based Decision Making
- Introductory Machine Learning Concepts
- AI Applications, Ethics & Responsible AI

Internship Profile

Structured learning framework provided by Kodacy for building a strong foundation in AI.



Structured Learning Experience

The internship offered a well-structured curriculum introducing the core ideas of Artificial Intelligence — from intelligent agents and search-based problem solving to knowledge representation, basic learning models, and real-world AI applications.

Through guided modules and conceptual case studies, the programme connected classical AI theory with the way modern intelligent systems are designed and reasoned about.



Artificial Intelligence

Intelligent Systems



Search & Reasoning

Problem Solving



Probability & Decisions

Bayesian Thinking



Intro to Machine Learning

Pattern Recognition



AI Applications

NLP, Vision & Chatbots

Core Learning Objectives

The primary technical pillars established for the Introduction to AI internship.

1

AI Foundations

Studying the history, definitions, classifications, and real-world scope of Artificial Intelligence.

2

Agents & Search

Understanding intelligent agents, state-space representation, and heuristic search strategies.

3

Knowledge & Logic

Exploring knowledge representation, inference, and rule-based expert systems.

4

Learning & Ethics

Introducing basic ML concepts alongside AI applications and responsible AI practice.

What is Artificial Intelligence?

Defining intelligence in machines, its evolution, and the two ways AI is classified.

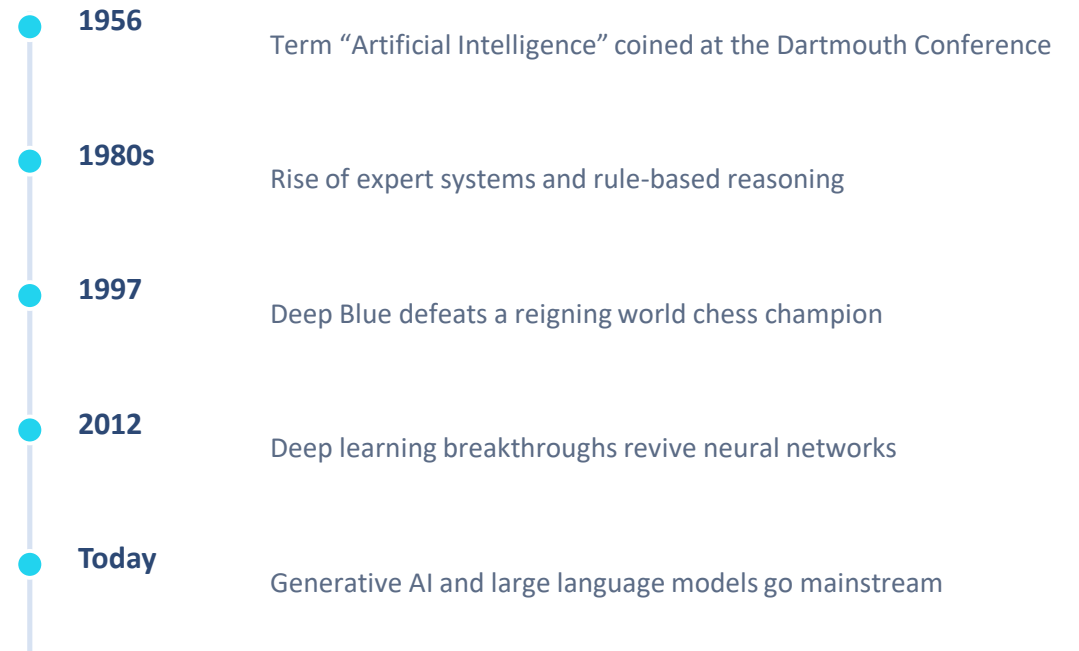
Definition

Artificial Intelligence is the branch of computer science concerned with building systems that can perceive, reason, learn, and act to achieve goals — tasks that would normally require human intelligence.

Narrow AI vs. General AI (AGI)

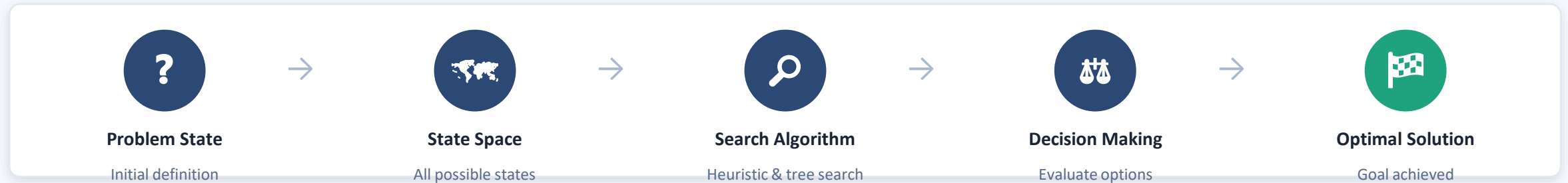
Narrow AI performs a specific task well (voice assistants, recommendation engines). General AI, still theoretical, would match human-level reasoning across any domain.

A Brief History of AI



AI Problem Solving: Agents & Search

How an intelligent agent perceives a problem and searches for an optimal solution.



Types of Intelligent Agents

- Simple Reflex Agents — act only on the current percept
- Goal-Based Agents — act to achieve a defined goal
- Utility-Based Agents — choose the best outcome
- Learning Agents — improve performance over time



Search & Decision Techniques

- State-space representation & graph traversal
- Minimax algorithm for adversarial games
- Probability, conditional likelihood & Bayes' Theorem
- Naive Bayes for simple classification tasks

Knowledge Representation & Reasoning

How AI systems store, structure, and reason over what they know.



Knowledge Representation

Semantic networks, frames, and structured schemes for storing facts



Logical Reasoning

Propositional and rule-based logic used to draw valid conclusions



Expert Systems

Rule-based inference engines that mimic domain-expert decision making



Ethical & Responsible AI

Early exposure to fairness, transparency, and accountability in AI design

Introduction to Machine Learning

A brief look at how machines learn patterns from data — the natural next step after classical AI.



Learning Paradigms

- Supervised Learning — learns from labeled examples
- Unsupervised Learning — finds hidden patterns
- Reinforcement Learning — learns via trial and reward
- Linear & Logistic Regression as starting models



Why It Matters for AI

- Gives AI systems the ability to improve with experience
- Powers modern applications like recommendations & vision
- Foundation for deep learning and neural networks
- A natural next step after classical, rule-based AI

Applications of Artificial Intelligence

Where the concepts studied show up in everyday intelligent systems.



Natural Language Processing

Chatbots, sentiment analysis, and language understanding



Computer Vision

Image classification, object detection & face recognition



Conversational AI

Rule-based & LLM-driven chatbot architectures



Healthcare & Industry

Diagnostics support, automation & assistive AI



Recommendation Systems

Personalized suggestions across e-commerce & media



AI Safety & Governance

Responsible deployment and ethical oversight

My AI Learning Journey

Sequential curriculum progression traversed throughout the internship at Kodacy.

01

AI Fundamentals

History, definitions & scope

02

Intelligent Agents

Agent types & environments

03

Search Strategies

State space & heuristics

04

Probability & Bayes

Statistical decision making

05

Knowledge & Logic

Representation & reasoning

06

Expert Systems

Rule-based inference

07

Intro to ML

Learning paradigms

08

Applications & Ethics

Real-world AI & governance

Skills & Knowledge Developed

Summary of conceptual understanding and analytical skills gained across the internship.



AI Foundations

Understanding of intelligent systems, history and the narrow-vs-general AI divide.



Search & Problem Solving

State-space search, heuristics, and adversarial game strategies.



Probability & Bayes

Conditional probability, Bayes' theorem, and simple probabilistic classifiers.



Knowledge & Logic

Representation schemes, logical inference, and expert-system design.



ML Fundamentals

Supervised, unsupervised and reinforcement learning at a conceptual level.



Ethics & Applications

Awareness of responsible AI practice across NLP, vision and automation.

Internship Experience & Conclusion

Reflective summary of academic growth, practical insight, and future learning pathways.



Key Internship Reflections

The Introduction to AI virtual internship provided a structured, disciplined introduction to how intelligent systems are designed and reasoned about.

Across the programme, I explored agent-based problem solving, search and probability, knowledge representation, and a first look at how machines learn from data — alongside the ethical questions modern AI raises.

Academic Value: *Built a strong conceptual foundation in AI for future coursework and applied ML projects.*



Future Learning Pathways



Applied Machine Learning

Hands-on model building using Python & scikit-learn.



Deep Learning Foundations

Neural networks, CNNs and RNNs for vision and language.



Advanced & Generative AI

Prompt engineering, LLMs and responsible deployment.

Certificate of Completion

Formal certificate awarded by Kodacy under the AICTE Virtual Internship Programme.



Credential Details

Awarded To: **Ayush Jha**

Organization: **Kodacy**

Domain: **Introduction to AI**

Program: **AICTE Virtual Internship**

Status: **Successfully Completed**

Kodacy

Certificate OF COMPLETION

This certificate is presented to:

Ayush Jha

for successfully completing the course on
Introduction to AI



Date Of Completion : September 10, 2028
Certificate ID : a1d844f92fa5942c
← Scan this QR to check validity


Authorized Signature
Akash Joseph (CEO)



Thank You!

Thank you for your time and guidance. Open to questions and academic evaluation.

AYUSH JHA

Introduction to AI Intern

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